A Mechanistic Interpretability Curriculum for Neil: From His Own GPT to a VLA Workshop Paper

TL;DR

- **Start by dissecting Neil's own char-level model for ~2 weeks, then move deliberately to TransformerLens + GPT-2 small**, because the documented, replicable circuits (induction heads, IOI, the standard scoring metrics) live in token-level models — but do the early hands-on dissection on his own model first, since that is exactly the "learn by touching the phenomenon" approach Neel Nanda and ARENA both prescribe.
- **The modern consensus path is: ARENA Chapter 1 (Transformer Interpretability), prioritized ruthlessly, plus just-in-time reading of A Mathematical Framework and Induction Heads — then pivot to research mini-projects within ~1 month**, not exhaustive reading. SAEs/superposition should be a skim-only "know what they are and their limits" item, not a deep investment, given the 2025-2026 field-wide cooling on SAEs.
- **The 28-checkpoint developmental series is a genuinely good, somewhat novel pedagogical exercise** that doubles as the seed of the eventual paper; the bridge to VLA interpretability runs through ViT/CLIP interpretability (Prisma) and the 2025 OpenVLA/π0 steering work, and a realistic NeurIPS/ICML workshop target exists (the Mechanistic Interpretability Workshop, plus actionable-interp and interpretable-AI workshops).

Key Findings

1. The current (2025-2026) consensus learning path is "minimum viable theory, then research," and ARENA is the spine of it. Neel Nanda's September 2025 guide "How To Become A Mechanistic Interpretability Researcher" is explicit: learn the ropes in ≤1 month (breadth-first), then move to 1-5 day research mini-projects, then 1-2 week sprints. He names ARENA Chapter 1 as the coding backbone and says to prioritize ruthlessly: Chapter 1.2 (Interpretability Basics — first 3 sections on tooling, direct observation, patching) is *essential*; 1.4.1 (Causal Interventions & Activation Patching) is *recommended*; 1.3.2 (SAEs) is *worthwhile but skim/skip section 1*. His repeated refrain: "Do not just read things. Mech interp is a fundamentally empirical science." This maps perfectly onto Neil's "build first, read after touching the phenomenon" style.

2. ARENA is the right curriculum, and it's now a living resource. ARENA (the Alignment Research Engineer Accelerator), curriculum by Callum McDougall, is organized into chapters each requiring roughly a week of full-time study. The repo is now stabilized as "ARENA 3.0" (kept at that name for backwards compatibility; it's the latest version and won't be replaced). Chapter 1 covers building transformers from scratch, then mech interp: TransformerLens, induction heads, the IOI circuit in GPT-2 small, activation/path patching, SAEs, and toy-model case studies (balanced bracket classifier, grokking/modular arithmetic with Fourier analysis, OthelloGPT world models). Only the first two exercise sets (build/train transformer + intro mech interp incl. induction heads & TransformerLens) are compulsory; the rest are pick-your-own with no prerequisites. GitHub + 2

3. Core primary sources, read just-in-time: Elhage et al., "A Mathematical Framework for Transformer Circuits" (2021, attention-only 2-layer models, QK/OV circuits, composition); Olsson et al., "In-context Learning and Induction Heads" (2022); nostalgebraist's "logit lens" (2020); Wang et al., "Interpretability in the Wild" (IOI circuit in GPT-2 small); and the 2025 field-state review Sharkey et al., "Open Problems in Mechanistic Interpretability" (arXiv:2501.16496, ~30 authors, good for landscape but "a bunch of opinionated and disagreeable researchers writing their own sections," per Nanda — don't over-defer).

4. What forms in a 6-layer char-level Shakespeare model — managed expectations. Per Olsson et al., induction heads form in models of *any size provided they have more than one layer*, at a sharp phase change visible as a bump in the loss curve; induction is an *algorithm over embeddings, not memorized n-gram statistics*. Neil's model has 6 layers, so it clears the structural bar. Two important caveats: (a) the *strong causal* evidence in Olsson et al. was for attention-only models; for models *with MLPs* (like Neil's) the evidence is *correlational* — so claims about "the induction circuit" in his model are hypotheses to test, not givens. (b) Char-level induction is expected to be **weaker and fuzzier than BPE-token induction**: a single character is a far more ambiguous prefix-match key than a BPE token, so useful char-level induction tends to operate over *multi-character prefixes* — effectively in-context copying of repeated strings like speaker names ("ROMEO:", "JULIET:"). Reliably findable in tiny char models: previous-token heads, fixed-offset positional heads, position-0 "rest/attention-sink" behavior, bigram statistics (learned first), skip-trigrams, and copying of repeated names. There is **no published circuit-level reverse-engineering of this exact model**, which is precisely why it's a good exercise — and a loadable equivalent checkpoint exists (`sosier/nanoGPT-shakespeare-char-tied-weights`, confirmed 10.65M params, vocab 65, 6 blocks). Transformer Circuits + 3

5. Tooling progression: raw hooks briefly, then TransformerLens — and TransformerLens can ingest his model. The consensus is that `register_forward_hook` in raw PyTorch is worth doing *once* to demystify what a hook is, then move to TransformerLens for the short feedback loops. TransformerLens supports custom models via `HookedTransformerConfig` + `HookedTransformer.from_config(cfg)`, and critically it **ships a nanoGPT weight-conversion module** (`transformer_lens.pretrained.weight_conversions.nanogpt`), so Neil can port his exact architecture and his 28 checkpoints into the HookedTransformer format and get clean, named activations (with LayerNorm folding, weight centering). As of Sept 2025 TransformerLens v3 is in alpha. Alternatives: **nnsight** (wraps HuggingFace models, better for large models / remote execution via NDIF), **pyvene** (declarative causal interventions), **SAELens** (training/using SAEs), and **ViT-Prisma** (the "TransformerLens for vision," Hooked Vision Transformers + pretrained CLIP/DINO SAEs). BorisTheBrave + 2

6. The developmental angle is real and active. "Developmental interpretability" (devinterp), grounded in singular learning theory (Watanabe; Lau et al.'s local learning coefficient; Hoogland et al. on developmental stages), studies how structure emerges through training phase transitions, motivated explicitly by Olsson et al.'s induction phase change and Elhage et al.'s toy superposition models. Tigges, Hanna, Yu & Biderman (NeurIPS 2024, arXiv:2407.10827) tracked circuits across 300B training tokens in Pythia models from 70M–2.8B params and found that "task abilities and the functional components that support them emerge consistently at similar token counts across scale...although such components may be implemented by different attention heads over time, the overarching algorithm that they implement remains." Standard metrics to track across checkpoints: **prefix-matching score** (Olsson et al.: average attention from a token to the token following its previous occurrence on a repeated random sequence), **induction score** (TransformerLens convention: attention 49-back on a 50-token repeated sequence), **previous-token matching score**, plus loss-curve derivatives and attention entropy. Pythia and Mistral provide public checkpoint series; Neil's 28-checkpoint series (dense early: 0,10,25,50,100,150,200,250,350,500,750 then every 250 to 4999) is well-designed to catch an early phase change. arxiv + 4

7. The path to VLA interpretability has concrete stepping stones and a real 2025-2026 literature. The bridge: GPT-2 interp → ViT/CLIP interp (ViT-Prisma, logit lens on images, attention-head viz, activation patching that flips a class prediction; the documented CLIP head L22.H6 in ViT-L/14, "which has been previously reported to strongly respond to images of animals" — Gandelsman, Efros & Steinhardt, "Interpreting CLIP's Image Representation via Text-Based Decomposition," ICLR 2024, arXiv:2310.05916) → multimodal/VLA. The seminal VLA mech-interp paper is Häon, Stocking, Chuang & Tomlin (UC Berkeley EECS), "Mechanistic Interpretability for Steering Vision-Language-Action Models" (CoRL 2025, PMLR v305:2743–2762, arXiv:2509.00328): "we introduce the first framework for interpreting and steering VLAs via their internal representations...We project feedforward activations within transformer layers onto the token embedding basis, identifying sparse semantic directions - such as speed and direction - that are causally linked to action selection." They steer OpenVLA (LIBERO sim) and π0 (physical UR5) zero-shot. Follow-ups already exist (e.g. arXiv:2603.19233, a mechanistic study across five VLAs incl. π0.5, OpenVLA-OFT, finding that mean-pooling across token positions destroys action-critical info and that causal validation requires rollout-based evaluation). What transfers from GPT-2-era skills: hooks, activation patching, logit/embedding-space projection, attention analysis, steering vectors.

What's different: heterogeneous token streams (vision+language+proprioception), and the need for *rollout-based* causal validation rather than logit inspection. arxiv

8. Workshop venues are real and recurring. Per mechinterpworkshop.com, "140+ papers submitted to our ICML 2024 workshop"; that CFP took short (4pp) and long (8pp) papers (deadline May 29 2024), offered $1,750 in best-paper prizes, and featured Chris Olah, Jacob Steinhardt, David Bau and Asma Ghandeharioun. The Mechanistic Interpretability Workshop then ran at NeurIPS 2025 (short ≤4pp / long ≤9pp papers; CFP closed Aug 22, 2025) and is slated for ICML 2026. There is also an "Actionable Interpretability" workshop and an "Interpretable AI: Past, Present and Future" workshop at NeurIPS, and a CVPR 2025 "Mechanistic Interpretability in Vision" (MIV) workshop where Prisma was an oral. These are realistic targets — Nanda explicitly notes people overestimate the workshop-paper bar. ResearchGate

9. Pitfalls and what to skip (from Nanda, Olah, Conmy). The dominant failure modes for self-taught learners:

- **Setup/meta-work procrastination**: "A common mistake is to spend days to weeks setting up epic infrastructure, or writing a perfect, sophisticated project proposal. Jump in with some quick experiments... The vibe should be closer to a weekend hackathon, rather than a 2 month internship." Neel Nanda Neel Nanda
- **Reading instead of doing**: "a common mistake among academic types is to spend months reading as many papers as they can get their hands on before writing code. Don't do it." alignmentforum
- **Missing the simple explanation**: "The most common mistake in junior MI researchers is coming up with an elaborate explanation for a seemingly complex behaviour and missing a simple explanation." Familiarize with skip-trigrams/bigrams/copying/induction from A Mathematical Framework and *always check whether the behavior could be done with some combination of these* before invoking something fancy. Neel Nanda Neel Nanda
- **"Excitement is evidence of bullshit"**: exciting results are more likely false; red-team relentlessly, every result is false until proven otherwise.
- **Fads to deprioritize** (Nanda's explicit list): interpreting toy models on algorithmic tasks ("we basically know that sometimes models trained on algorithmic tasks are interpretable"); pure circuit-identification via causal interventions where there's no "what next?"; and *incremental SAE research* — "We're at the tail end of a fad of incremental sparse autoencoder research... I do not think they were a gamechanger in the way that I hoped they might be." alignmentforum alignmentforum

10. SAEs in 2026: know them, don't over-invest. Multiple 2025 results have cooled the field on SAEs: Kantamneni, Engels, Rajamanoharan, Tegmark & Nanda (ICML 2025, arXiv:2502.16681) tested SAE probes on 113 binary-classification datasets across four regimes (data scarcity, class imbalance, label noise, covariate shift) and found "SAE probes underperform the baseline of logistic regression in each regime when taking the mean across datasets." The GDM mech interp team (Smith et al. 2025) argued SAEs won't be a gamechanger and the field is over-invested; and Heap, Lawson, Farnik & Aitchison (arXiv:2501.17727, "Sparse Autoencoders Can Interpret Randomly Initialized Transformers") found that "random and trained transformers produce similarly interpretable SAE latents...Further, we find that SAE quality metrics are broadly similar for random and trained transformers" — the "dead salmon" result. Feature splitting, absorption, and non-uniqueness across random seeds are documented problems. The pragmatic 2026 consensus (Nanda's "A Pragmatic Vision for Interpretability," Dec 2025): ambitious full reverse-engineering "seems basically doomed"; the action is in downstream tasks, model organisms, auditing games, linear probes for monitoring, and attribution-graph circuit analysis (transcoders, Neuronpedia). arXiv

Details: The Ordered Roadmap

Assume roughly 3-4 focused months, part-time within a broader research effort. Week counts are for a motivated part-timer; compress if full-time.

Phase 0 — Tooling bootstrap (3-5 days, do NOT let this expand)

This is the single highest procrastination risk for Neil. Time-box it hard.

- On the **M4 Pro (MPS)**: set `PYTORCH_ENABLE_MPS_FALLBACK=1` (some ops aren't implemented in MPS and will otherwise crash); know that **MPS does not support float64** (interpretability code that wants double precision must run on CPU); watch for CPU-vs-MPS device-mismatch errors (`all input tensors must be on the same device`) and the fact that MPS kernels are newer and occasionally buggy vs CPU/CUDA. For small models like his, MPS is fine and the 24GB unified memory is plenty. Hugging Face + 3
- Do **one** raw-PyTorch `register_forward_hook` exercise: grab the residual stream at a layer, confirm shapes, cache it. This demystifies hooks. Then stop using raw hooks.
- Install TransformerLens. Port his model via `HookedTransformerConfig` and the nanoGPT weight-conversion path; verify logits match his original model on a few prompts (this catch-bugs step matters — a subtly wrong port silently corrupts everything downstream).
- **When the 4080 matters**: training (he's done it) and later any SAE training or GPT-2-medium-and-up work. For Phase 1-2 dissection of a 10.8M model, the Mac is entirely sufficient. Use the 4080 PC for batched checkpoint sweeps and CUDA-only library features.

Phase 1 — Dissect his own model (~2 weeks)

Goal: touch every core technique on a model he understands completely. Exercises in order, each with what to measure:

1. **Attention pattern visualization** (day 1-2). Use CircuitsVis (`cv.attention.attention_patterns`) on Shakespeare text. Measure: which heads attend to position 0 (rest/sink behavior — expect several "idle" heads showing a vertical stripe); which attend to fixed offsets (previous-token heads ≈ attention to t-1); which attend to repeated characters/names. *Just-in-time read*: skim A Mathematical Framework §"attention heads as information movement." Learnmechinterp
2. **Direct logit attribution / logit lens** (day 3-4). Project intermediate residual stream through the unembedding. Measure: at which layer does the prediction for the next character "resolve"? Are common characters (space, 'e') resolved earlier? *Read*: nostalgebraist's logit lens post.
3. **Previous-token-head and bigram check** (day 5-6). Confirm bigram statistics are present early (after 't' → 'h'); identify previous-token heads quantitatively (previous-token matching score). This grounds the "check the simple explanation first" discipline. Netfuture
4. **Induction probe** (day 7-9). Feed a repeated random-character sequence; compute the **induction score** (attention 49-back on a 50-char repeat) and **prefix-matching score** for every (layer, head). *Expectation management*: he may find weak/fuzzy induction operating over multi-character prefixes (name copying) rather than crisp single-char induction — document whatever he finds honestly. *Read*: Olsson et al. (the six arguments + the phase-change section). arxiv
5. **Ablation** (day 10-11). Zero-ablate candidate heads; measure loss change on held-out Shakespeare and on the synthetic repeat task. Which heads matter for copying vs general LM loss?
6. **Activation patching** (day 12-14). Build a minimal clean/corrupted pair (e.g., a speaker name that should be copied vs. a different name) and patch residual-stream/head outputs to localize where the "which name" information travels. *Read*: ARENA 1.4.1; the "How-to Transformer MI in 50 lines" walkthrough.

Throughout: keep a research log; make a new plot every few minutes when experiments are fast (Nanda's pace benchmark).

Phase 2 — TransformerLens + GPT-2 small, documented circuits (~3 weeks)

Switch to token-level models where ground truth is documented, so he can check his technique against known answers.

- Work ARENA 1.2 (essential) and 1.4.1 (patching) on GPT-2 small. Replicate the induction-head finding on `gpt2-small` / a 2-layer attention-only toy model, then the **IOI circuit** (Wang et al.) — this is the canonical end-to-end circuit exercise and teaches path patching.
- Why this order: his own char model gives intuition and ownership; GPT-2 gives *correctness checks* and the shared vocabulary of the field. Doing his model first means he arrives at IOI already fluent in hooks, patching, and head-scoring.
- *Read just-in-time*: Wang et al. IOI; skim Conmy et al. (ACDC) for automated circuit discovery.

Phase 3 — Developmental study of his 28 checkpoints (~3 weeks) — the paper seed

This is where his unique asset pays off and where a workshop contribution can germinate.

- Port all 28 checkpoints into HookedTransformer (batch this on the 4080).
- For each checkpoint, compute and plot over training: induction score and prefix-matching score (per head and max-over-heads), previous-token matching score, attention entropy per head, and the loss curve + its discrete derivative. Overlay them.
- Hypotheses to test: Does a phase change show up in his dense early checkpoints (the 0-750 cluster is well-placed to catch it)? Does a previous-token head form *before* the induction/name-copying head (the predicted two-stage composition)? Does any loss-derivative bump coincide?
- *Read just-in-time*: "Towards Developmental Interpretability"; Tigges et al. (2024, arXiv:2407.10827); optionally the devinterp/SLT "Loss Landscape Degeneracy and Stagewise Development in Transformers" (Hoogland et al., arXiv:2402.02364) for the local-learning-coefficient metric and its checkpoint methodology.
- **Caveat to flag in any writeup**: char-level + MLP-containing means evidence is correlational; he should use ablations to add causal weight, and compare against the documented token-level phase-change timing rather than claiming novelty without that baseline.

Phase 4 — Introduce SAEs/causal methods lightly (~1 week)

- Do ARENA 1.3.2 at skim depth: what an SAE is, its strengths/weaknesses, how to *use* a pretrained one (Neuronpedia). Do **not** train SAEs from scratch or build an SAE-centric project.
- Optionally make a steering vector (Nanda's happiness-steering exercise: generate ~32 happy / 32 sad prompts via an LLM API, take the mean activation difference at a mid layer, add it during generation) to learn activation addition — directly transferable to VLA steering later.
- Explore one attribution graph on Gemma-2-2b via Neuronpedia to see the current frontier method.

Phase 5 — Bridge to vision and VLA (~4-5 weeks)

- **ViT/CLIP interp** (~2 weeks): install ViT-Prisma; run the Main ViT Demo (logit lens on images, attention-head viz, activation patching that flips tabby-cat→Border-collie). Reproduce or inspect the documented CLIP animal head (L22.H6 in ViT-L/14; Gandelsman et al. 2024). Note the key vision-vs-language difference Prisma documents: CLIP internal representations are *much less sparse* than language (CLIP-B/32 needs ~500+ L0 per patch), so vision SAEs behave differently. GitHub The Moonlight
- **VLA** (~2-3 weeks): read Häon et al. (2509.00328) closely and the multi-VLA follow-up (2603.19233). Reproduce the FFN-projection / activation-steering method on OpenVLA in LIBERO (the open, simulation-only path — no robot needed; runs on the 4080). Pick a

concrete, *small* extension: e.g., apply his developmental-checkpoint lens to a VLA fine-tune, or test whether the "semantic direction" steering survives rigorous baselines/ablations (the kind of rigor Nanda emphasizes), or extend the "not all features are equal" rollout-validation finding.
- Target a workshop paper for the Mechanistic Interpretability Workshop (ICML 2026) or an actionable-interpretability/MIV venue, short-paper format.

Recommendations

Staged next steps with decision thresholds:

1. **This week**: Time-box Phase 0 to 5 days. *Threshold to proceed*: his model's logits match between native and HookedTransformer versions on 5 test prompts. If porting drags past a week, fall back to doing Phase 1 with raw hooks on the native model and port later — do not let tooling block science.
2. **Weeks 2-3 (Phase 1)**: Complete all six dissection exercises on his own model. *Threshold*: he can compute an induction score and an activation-patching result unaided. *Red flag*: if he's writing philosophy about "what attention means" instead of generating plots, stop and run the next experiment.
3. **Weeks 4-6 (Phase 2)**: Replicate induction + IOI on GPT-2 small. *Threshold to advance*: he reproduces the IOI name-mover/duplicate-token head story and his numbers roughly match the paper.
4. **Weeks 7-9 (Phase 3)**: The checkpoint study. *Threshold that this becomes the paper*: he finds a clean, reproducible developmental signal (a phase change, or a clear before/after ordering of head formation) that he can state in one sentence and defend with an ablation. *If no clean signal by end of week 9*: pivot — write it up as a negative/methods result (Nanda explicitly values well-analyzed negative results) and move to Phase 5.
5. **Weeks 10-11 (Phase 4)**: SAE/steering skim. Keep it to a week.
6. **Weeks 12-16 (Phase 5)**: Vision then VLA. *Threshold for paper submission*: a reproducible result with a real baseline and at least one ablation, framed against an existing VLA-interp paper.

Cross-cutting recommendations:

- **Use LLMs aggressively as tutors** (Nanda's strong advice): generate exercises from A Mathematical Framework, summarize papers, but write your own experiment code first when learning a library (Cursor for throwaway work, browser-LLM-as-tutor for ARENA).
- **Find a mentor / collaborators early**: MATS (Nanda's stream), the Open Source Mechanistic Interpretability Slack, the EleutherAI Discord. A weekly check-in dramatically accelerates the slow skills (prioritization, taste). For a VLA paper specifically, reaching out to first authors (not famous last authors) of the Häon et al. line is high-value.

- **Default to publishing**: a blog post or workshop paper is the best credential and the best forcing function.

Benchmarks that would change the plan:

- If he discovers strong, crisp char-level induction in Phase 1 (contrary to the weak/fuzzy expectation), that itself is a publishable curiosity — lean into it.
- If the checkpoint study shows nothing by week 9, pivot to VLA sooner.
- If VLA reproduction proves too heavy on the 4080, stay in CLIP/ViT-Prisma interpretability for the paper — it's a legitimate workshop venue (MIV) and a cleaner bridge.

Caveats

- **No published reverse-engineering of Neil's exact model exists**, so Phase 1/3 findings are genuinely his to discover — but that also means he has no ground-truth answer key for his own model (unlike GPT-2). This is why Phase 2 (known circuits) is non-negotiable as a technique check.
- **Char-level + MLP means correlational evidence**: any "induction circuit" claim about his model is a hypothesis requiring ablation/patching to support causally. The clean causal story in A Mathematical Framework was for attention-only models.
- **SAE guidance reflects a contested, fast-moving consensus.** The field's cooling on SAEs is real and well-sourced, but some researchers disagree; the recommendation to skim-not-invest is a judgment call optimized for Neil's time budget and "build-first" style, not a universal verdict.
- **TransformerLens v3 is in alpha** (as of late 2025); expect some API churn and possible MPS rough edges. The nanoGPT conversion path exists but verify numerically.
- **Time estimates assume part-time focus**; the broader research effort Neil is embedded in will compete for hours. The ≤1-month "learn the ropes" target is Nanda's for full-time work — Neil's calendar version is longer, which is fine.
- **VLA interpretability is a young subfield** (the seminal paper is from late 2025). That means low-hanging fruit but also shifting baselines and few established methods — a workshop paper is realistic; a top-conference paper in a few months is not the right expectation.